

Q.1 Radio waves are propagated in _____ whenever they are transmitted by an antenna.

- A. Right direction
- B. Left direction
- C. Only up and down direction

D. All directions

Q.2 In optical fiber cable, the ratio of the diameter of their core to the diameter of their cladding, both expressed in _____.

- A. Centimeters
- B. Millimeters
- C. Meters

D. Micrometers

Q.3 In circuit switching, when two end devices "A" and "B" connects with each other they send _____ request.

- A. Inquiry
- B. Switching
- C. Acceptance

D. Setup

Q.4 Packet Switching operates on _____ layer.

- A. Application layer
- B. Physical layer

C. Network layer

D. Data link layer

Q.5 Coaxial cable carries signals of higher frequency ranges than those in _____ cable.

A. Twisted pair

- B. Fiber optic
 - C. T Lines
 - D. Television
-

Q.6 Entire band in United States is regulated by _____ authorities.

- A. ANSI
- B. ITU
- C. NASA

D. FCC

Q.7 Congestion Control is a feature of _____ layer(s).

- A. All mentioned
 - B. Network
 - C. Transport**
 - D. Data Link
-

Q.8 In virtual circuit approach when a frame enters a switch and when it leaves a switch its VCI_____.

- A. Remain the Same
 - B. Always Changes
 - C. Sometimes Remain Same
 - D. Sometimes Changes**
-

Q.9 _____ category of coaxial cable is used for thin Ethernet.

- A. RG-1
 - B. RG-58**
 - C. RG-47
 - D. RG-59
-

Q.10 Cable TV networks use_____ cables.

- A. UTP
 - B. Coaxial**
 - C. Twisted pair
 - D. STP
-

Q.1 In circuit switching, when two end devices "A" and "B" connects with each other they send _____ request.

- A. Inquiry
 - B. Acceptance
 - C. Setup**
 - D. Switching
-

Q.2 _____ is the scope of VCI variable in virtual circuit approach.

- A. Variable
 - B. Private
 - C. Local**
 - D. Global
-

Q.3 _____ is responsible for governing node to node communication.

- A. Session Layer
 - B. Application Layer
 - C. Data Link layer**
 - D. Presentation Layer
-

Q.4 Virtual Circuit Identifier is the _____ of packet.

- A. Header**
 - B. Size
 - C. Length
 - D. Address
-

Q.5 The size of the packet is determined by the network and _____.

- A. Governing Protocol
 - B. Switching Type**
 - C. Instructions
 - D. Delays
-

Q.6 A twisted pair cable consists of _____ conductors.

- A. Five
 - B. Two**
 - C. Three
 - D. Four
-

Q.7 _____ are used for multicast communications, such as radio and television.

- A. Infrared waves
 - B. Radio waves**
 - C. Light wave
 - D. Microwaves
-

Q.8 Congestion Control is a feature of _____ layer(s).

- A. Transport
 - B. Network
 - C. All mentioned**
 - D. Data Link
-

Q.9 Coaxial cables are categorized by their _____ ratings.

- A. Electronic Industries Association (EIA)
 - B. Amplitude modulation (AM)
 - C. Radio Government (RG)**
 - D. Frequency modulation (FM)
-

Q.10 In optical fiber a glass or plastic core is surrounded by a _____.

- A. Inner Conductor shield
 - B. Cladding**
 - C. Outer conductor shield
 - D. Insulator
-

Q.1 Connection setup means using _____ channels in circuit switched network.

- A. Authenticated
 - B. Dedicated**
 - C. Satellite
 - D. Communication
-

Q.2 A local telephone network use _____ network.

- A. Circuit switched**
 - B. Bit switched
 - C. Packet switched
 - D. Line switched
-

Q.3 Packet Switching operates on _____ layer.

- A. Physical layer
 - B. Data link layer
 - C. Application layer
 - D. Network layer**
-

Q.4 In datagram network, each data packet contains its _____.

- A. Number
 - B. ID
 - C. Address**
 - D. Name
-

Q.5 Ships use _____ communication bands of frequency.

- A. 3-30 THz
 - B. 3-30 kHz
 - C. 3-30 MHz**
 - D. 3-30 GHz
-

Q.6 Coaxial cable carries signals of higher frequency ranges than those in _____ cable.

- A. Fiber optic
 - B. Television
 - C. Twisted pair**
 - D. T Lines
-

Q.7 _____ does not happen until all the data is transferred and all of the resources are freed.

- A. Switching
 - B. Resource Allocation
 - C. Tear Down**
 - D. Acknowledgement
-

Q.8 According to stats, fewer than _____ crossbar points can be used at once.

- A. 25%**
 - B. 100%
 - C. 30%
 - D. 14%
-

Q.9 Microwaves are _____.

- A. Multi directional
 - B. Bi directional
 - C. Omni directional
 - D. Unidirectional**
-

Q.10 In a circuit switched network, the link between the switches is segregated into ____ number of channels.

- A. n^2
 - B. $n-1$**
 - C. $2n$
 - D. n
-

Q.1 A virtual-circuit network operates on _____ layer.

- A. Physical
 - B. Datalink
 - C. Application
 - D. Network**
-

Q.2 Data link layer is divided into _____ sub layers.

- A. Seven
 - B. Two**
 - C. Three
 - D. Five
-

Q.3 Which one of the following is not among the required phases for the actual communication in a circuit-switched network?

- A. Connection teardown
 - B. Setup
 - C. Data transfer
 - D. Protocol**
-

Q.4 _____ category of coaxial cable is used for thin Ethernet.

- A. RG-59
 - B. RG-1
 - C. RG-58**
 - D. RG-47
-

Q.5 In circuit switching, when two end devices "A" and "B" connects with each other they send _____ request.

- A. Setup**
 - B. Switching
 - C. Inquiry
 - D. Acceptance
-

Q.6 Among the following cables given below, _____ cable has the highest cost as compared to others.

- A. STP
 - B. Coaxial cable
 - C. UTP
 - D. Fiber optical**
-

Q.7 _____ have established standards for using these signals for communication between devices such as keyboards, mice, PCs, and printers.

- A. Radio waves
 - B. Infrared waves**
 - C. Microwaves
 - D. Simple waves
-

Q.8 Some manufacturers provide a special port called the _____ port that allows a wireless keyboard to communicate with a PC.

- A. Consol
 - B. IrDA**
 - C. USB
 - D. Parallel
-

Q.9 In virtual Circuit network all packets follow _____ path.

- A. Individual
 - B. Centralized
 - C. Dedicated**
 - D. Alternative
-

Q.10 In cross bar switch, the cross points can be calculated by____ inputs and outputs.

- A. Multiplying**
 - B. Subtracting
 - C. Dividing
 - D. Adding
-

Q.1 A virtual-circuit network is a cross between a _____ network and a datagram network.

- A. Both message and packet switched network
 - B. Circuit switched**
 - C. Message switched
 - D. Packet switched
-

Q.2 RG-11 having impedance of 50 Ohm is used in _____ Ethernet.

- A. Thin**
 - B. Thick
 - C. Gigabit
 - D. Fast
-

Q.3 A twisted pair cable consists of _____ conductors.

- A. Two**
 - B. Three
 - C. Four
 - D. Five
-

Q.4 _____ and _____ are the two types of addressing in virtual circuit approach.

- A. Private, Public
 - B. Dependent, Independent
 - C. Variable, Dynamic
 - D. Local, Global**
-

Q.5 Density of the core at the destination decreases in _____ propagation mode.

- A. Single mode, step index
 - B. Multimode, step index**
 - C. Multimode, graded index
 - D. Single mode, graded index
-

Q.6 The material used for conduction in twisted pair cable is _____.

- A. Aluminum
 - B. Silver
 - C. Gold
 - D. Copper**
-

Q.7 Frequency division multiplexing (FDM) or ____ are the resources that a station reserves before starting communication.

- A. ADM (Amplitude Division Multiplexing)
 - B. SDM (Space Division Multiplexing)
 - C. All of the given
 - D. TDM (Time Division Multiplexing)**
-

Q.8 _____ does not happen until all the data is transferred and all of the resources are freed.

- A. Switching
 - B. Resource Allocation
 - C. Tear Down**
 - D. Acknowledgement
-

Q.9 DLC in Data Link Layer stands for ____.

- A. Data Link Control**
 - B. Divide Line Communication
 - C. Demand Link Coordination
 - D. Data Line Code
-

Q.10 Electromagnetic waves ranging in frequencies between 1 and 300 GHz are called_____.

- A. Infrared waves
 - B. Microwaves**
 - C. Light wave
 - D. Radio waves
-

Q.1 Coaxial cable carries signals of higher frequency ranges than those in _____ cable.

- A. T Lines
 - B. Twisted pair**
 - C. Fiber optic
 - D. Television
-

Q.2 LF (Low Frequency) band is used in ____ propagation.

- A. Sky
 - B. Single Mode
 - C. Line of Sight
 - D. Ground**
-

Q.3 In datagram network, each data packet contains its ____.

- A. Address**
 - B. Name
 - C. ID
 - D. Number
-

Q.4 In _____ switching resources are allocated during the entire duration of the connection and these resources are unavailable to other connections.

- A. Circuit Switched Networks**
 - B. Both circuit and message switch network.
 - C. Packet-switched networks
 - D. Message switched networks
-

Q.5 In circuit switched networks we have low efficiency but minimal _____.

- A. Throughput
 - B. Errors
 - C. Delay**
 - D. Speed
-

Q.6 RG-11 having impedance of 50 Ohm is used in _____ Ethernet.

- A. Fast
 - B. Thin
 - C. Gigabit
 - D. Thick**
-

Q.7 The size of the packet is determined by the network and _____.

- A. Instructions
 - B. Switching Type
 - C. Governing Protocol**
 - D. Delays
-

Q.8 A virtual-circuit network operates on _____ layer.

- A. Application
 - B. Network**
 - C. Datalink
 - D. Physical
-

Q.9 _____ and _____ are the two types of addressing in virtual circuit approach.

- A. Dependent, Independent**
 - B. Local, Global
 - C. Variable, Dynamic
 - D. Private, Public
-

Q.10 In a circuit switched network, the link between the switches is segregated into ____ number of channels.

- A. $2n$
 - B. n^2
 - C. $n-1$
 - D. n**
-

Q.1 VCI address is _____ address in virtual circuit approach.

- A. Global
 - B. Local**
 - C. Variable
 - D. Private
-

Q.2 Datagram approach and virtual circuit approach are two popular approaches which lead us to _____.

- A. Circuit switching
 - B. Line switching
 - C. Packet switching**
 - D. Dynamic switching
-

Q.3 Cable TV networks use _____ cables.

- A. Coaxial**
 - B. Twisted pair
 - C. STP
 - D. UTP
-

Q.4 Ships use _____ communication bands of frequency.

- A. 3-30 MHz**
 - B. 3-30 THz
 - C. 3-30 kHz
 - D. 3-30 GHz
-

Q.5 Data Link layer receives services from the _____ layer.

- A. Session
 - B. Application
 - C. Physical**
 - D. Transport
-

Q.6 Among the following cables given below, _____ cable provides highest bandwidth and less attenuation.

- A. Fiber optical**
 - B. Coaxial cable
 - C. STP
 - D. UTP
-

Q.7 LF (Low Frequency) band is used in ____ propagation.

- A. Line of Sight
 - B. Single Mode
 - C. Sky
 - D. Ground**
-

Q.8 Virtual-Circuit Networks and datagram networks are the sub categories of _____.

- A. Circuit Switched Networks
 - B. Message switched networks
 - C. Both circuit and message switched networks
 - D. Packet-switched networks**
-

Q.9 Data Link layer provides services to the _____ layer.

- A. Transport
 - B. Session
 - C. Application
 - D. Network**
-

Q.10 A virtual-circuit network operates on _____ layer.

- A. Application
 - B. Physical
 - C. Datalink
 - D. Network**
-

Q.1 In a circuit switched network, the link between the switches is segregated into ____ number of channels.

- A. $n-1$
 - B. $2n$
 - C. n**
 - D. n^2
-

Q.2 ____ has less electromagnetic interference as compared to other guided medium.

- A. Unshielded Twisted Pair
 - B. Fiber Optic**
 - C. Coaxial
 - D. Shielded Twisted Pair
-

Q.3 Radio waves are propagated in _____ whenever they are transmitted by an antenna.

- A. Only up and down direction
 - B. Left direction
 - C. All directions**
 - D. Right direction
-

Q.4 Multi-mode technology has _____ main types for light propagation.

- A. Two**
 - B. Three
 - C. Four
 - D. Five
-

Q.5 Datagram approach and virtual circuit approach are two popular approaches which lead us to _____.

- A. Dynamic switching
 - B. Packet switching**
 - C. Circuit switching
 - D. Line switching
-

Q.6 The fiber optic cable of type 7/125 is of ____ mode.

- A. Single mode, step index**
 - B. Single mode, graded index
 - C. Multimode, step index
 - D. Multimode, graded index
-

Q.7 Switching in Datagram is normally done at_____.

A. Network Layer

B. Physical Layer

C. Transport Layer

D. Session Layer

Q.8 Connection setup means using _____channels in circuit switched network.

A. Satellite

B. Dedicated

C. Communication

D. Authenticated

Q.9 Electromagnetic waves ranging in frequencies between 1 and 300 GHz are called_____.

A. Infrared waves

B. Microwaves

C. Light wave

D. Radio waves

Q.10 _____ does not happen until all the data is transferred and all of the resources are freed.

A. Acknowledgement

B. Resource Allocation

C. Tear Down

D. Switching

Q.1 Cable TV networks use _____ cables.

- A. STP
 - B. Coaxial**
 - C. Twisted pair
 - D. UTP
-

Q.2 Wireless transmission can be divided into _____ broad groups.

- A. Four
 - B. Two**
 - C. Three
 - D. Five
-

Q.3 Cellular telephone uses _____ waves for communication.

- A. Radio waves**
 - B. Light waves
 - C. Microwaves
 - D. Infrared waves
-

Q.4 _____ category of coaxial cable is used for thin Ethernet.

- A. RG-59
 - B. RG-47
 - C. RG-1
 - D. RG-58**
-

Q.5 Packet Switching operates on _____ layer.

- A. Data link layer
 - B. Network layer**
 - C. Application layer
 - D. Physical layer
-

Q.6 Most commonly used connector for twisted pair cable is _____.

- A. RJ-52
 - B. RJ-11
 - C. RJ-45**
 - D. RJ-44
-

Q.7 Media Access Control (MAC) and Data Link Control (DLC) constitute _____ layer.

- A. Network
 - B. Transport
 - C. Data link**
 - D. Physical
-

Q.8 _____ is the scope of VCI variable in virtual circuit approach.

- A. Variable
 - B. Global
 - C. Local**
 - D. Private
-

Q.9 Radio waves are propagated in _____ whenever they are transmitted by an antenna.

- A. All directions**
 - B. Left direction
 - C. Right direction
 - D. Only up and down direction
-

Q.10 Paging System in data communication uses _____ waves.

- A. Micro waves
 - B. Radio waves**
 - C. Ultraviolet waves
 - D. Infrared waves
-

Q.1 The _____ is the number of signal elements sent per unit time.

- A. Bit rate
 - B. Period
 - C. Signal rate**
 - D. Data rate
-

Q.2 The message 110 sent by a source is received by a destination as 111. This is _____ type of error.

- A. Burst
 - B. Multi-Bit
 - C. Single-Bit**
 - D. Byte
-

Q.3 _____ are used to exchange session management and control information between connected devices.

- A. N-frames
 - B. I-frames
 - C. S-frames**
 - D. U-frames
-

Q.4 _____ is the process of converting digital data to digital signals.

- A. Line coding**
 - B. CSMA/CD
 - C. CSMA/CA
 - D. Segmentation
-

Q.5 _____ is the purpose of the Cyclic Redundancy Check (CRC) method

- A. Counting the number of bits in a frame
 - B. Regulating data flow
 - C. Detecting and correcting errors**
 - D. Adding redundancy for encryption
-

Q.6 The _____ protocol uses both flow and error control.

- A. HDLC**
 - B. Stop-and-Wait
 - C. ICMP
 - D. TCP/IP
-

Q.7 The term "single-bit error" refer to _____

- A. interfering with the flow control process
 - B. changing 1 bit of a data unit**
 - C. changing multiple bits in a data unit
 - D. changing the shape of the signal
-

Q.8 Payload field of the PPP frame is of variable size and it can carry maximum data up to _____.

- A. 10 bytes
 - B. 500 bytes
 - C. 1500 bytes**
 - D. 100 bytes
-

Q.9 If the Hamming distance between sent and received code word is _____, then it shows that received data is corrupted.

- A. $\neq 0$**
 - B. $=1$
 - C. $= 0$
 - D. $\neq 1$
-

Q.10 The purpose of block coding is _____

- A. removing errors from transmitted data
 - B. creating relationships between bits
 - C. adding redundancy to data in fixed-size blocks**
 - D. regulating data flow
-

Q.1 In _____ type of error, two or more bits in the data unit will be changed.

- A. Mono
 - B. Burst**
 - C. Single bit
 - D. Zero
-

Q.2 _____ is not a function of Data Link Control.

- A. Modulation**
 - B. Error Control
 - C. Framing
 - D. Flow Control
-

Q.3 The primary cause of unpredictable changes in bits during data transmission is _____

- A. decapsulation
 - B. framing
 - C. interference**
 - D. encryption
-

Q.4 High-Level Data Link Control (HDLC) defines _____ types of frames to deal with the transfer modes.

- A. Two
 - B. Four
 - C. Three**
 - D. Six
-

Q.5 In Block coding we divide our message into blocks of k bits called as _____

- A. Code words
 - B. Data words**
 - C. Passwords
 - D. Cross words
-

Q.6 Redundancy contribute to error detection by _____

- A. reducing the size of data frames
 - B. encrypting data for secure transmission
 - C. increasing the speed of data transmission
 - D. adding extra bits to create relationships for error detection**
-

Q.7 The term "single-bit error" refer to _____

- A. changing the shape of the signal
 - B. changing 1 bit of a data unit**
 - C. interfering with the flow control process
 - D. changing multiple bits in a data unit
-

Q.8 The extra bits added with the original data for error detection/correction are called _____.

- A. Fault bits
 - B. Process IDs
 - C. Port numbers
 - D. Redundant bits**
-

Q.9 Payload field of the PPP frame is of variable size and it can carry maximum data up to _____.

- A. 100 bytes
 - B. 500 bytes
 - C. 10 bytes
 - D. 1500 bytes**
-

Q.10 S-frames in High-Level Data Link Control (HDLC) are only used to transfer _____ information.

- A. User data
 - B. Original
 - C. Redundant
 - D. Control**
-

Q.1 In _____ protocol, the sender sends one frame at a time and waits for an acknowledgment before sending the next one.

- A. Simple
- B. Selective-Repeat
- C. Stop-and-Wait**
- D. Go-Back-N

Q.2 In digital communication, why do we use a checksum for error detection in a message?

- A. To encrypt the data
- B. To correct errors
- C. To facilitate faster data transmission
- D. To ensure even or odd parity**

Q.3 The extra bits added with the original data for error detection/correction are called _____.

- A. Port numbers
- B. Process IDs
- C. Redundant bits**
- D. Fault bits

Q.4 No matter whether the link is dedicated or broadcast, data link control (DLC) layer provides services between -----.

- A. Source and destination PCs
- B. Ethernet links
- C. Tow adjacent nodes**
- D. Virtual LANs

Q.5 Which of the following is a subset of Cyclic Codes?

- A. Parity Check
- B. Linear Block Codes
- C. Cyclic Redundancy Check (CRC)**
- D. Hamming Codes

Q.6 _____ is used in the Stop-and-Wait protocol to overcome the issue of duplication.

- A. Counter and Timer reset
- B. Acknowledgement numbering
- C. Frame sequencing and Acknowledgement numbering**
- D. Frame sequencing

Q.7 Error correction is more _____ than the error detection.

A. Difficult

B. Easy

C. Informal

D. Useless

Q.8 In even parity check code, the value of syndrome is _____ if the number of 1s is even.

A. 1

B. 4

C. 2

D. 0

Q.9 _____ is the process of converting digital data to digital signals.

A. Line coding

B. CSMA/CD

C. CSMA/CA

D. Segmentation

Q.10 In Block coding scheme, number of code words is always _____ data words.

A. Equal to

B. Greater than

C. Less than

D. Same as

Q.1 If the Hamming distance between sent and received code word is _____, then it shows that received data is corrupted.

- A. =1
- B. $\neq 1$
- C. $\neq 0$**
- D. = 0

Q.2 Redundancy contribute to error detection by _____

- A. encrypting data for secure transmission
- B. adding extra bits to create relationships for error detection**
- C. reducing the size of data frames
- D. increasing the speed of data transmission

Q.3 Go-Back-N protocol is one of the _____ layer protocols, which is used to control the flow of data during the transmission.

- A. Network
- B. Presentation
- C. Data link**
- D. Physical

Q.4 What is the primary function of a checksum in digital communication?

- A. Error detection**
- B. Error correction
- C. Encryption
- D. Data compression

Q.5 Hamming distance is important for error detection in coding as

- A. it measures the signal-to-noise ratio in communication.
- B. it indicates the strength of the error correction code.
- C. it measures the transmission speed of data.
- D. it represents the number of corrupted bits during transmission.**

Q.6 Stop-and-Wait protocol is used for _____.

- A. Flow control only
- B. Both flow and error control**
- C. Error control only
- D. Neither flow nor error control

Q.7 The message 1110 sent by a source is received by a destination as 1011. This is ____ type of error.

- A. Hamming
 - B. Uni-Bit
 - C. Burst**
 - D. Single-Bit
-

Q.8 In Linear Block codes, ____ of two valid code words creates another valid code word.

- A. XOR**
 - B. OR
 - C. NOR
 - D. AND
-

Q.9 CHAP is an Authentication Protocol, which uses _____ process to authenticate user information.

- A. Two-way
 - B. Four-way
 - C. Five-way
 - D. Three-way**
-

Q.1 The purpose of adding redundant bits in block coding is

- A. to provide error detection and correction capability**
 - B. to improve the efficiency of data transmission
 - C. to increase the size of data words
 - D. to decrease the size of codewords
-

Q.2 The purpose of a checksum in error detection is _____

- A. to verify the integrity of transmitted data**
 - B. to add redundancy for error correction
 - C. to regulate data flow between devices
 - D. to count the number of bits in a data frame
-

Q.3 The purpose of block coding is _____

- A. removing errors from transmitted data
 - B. creating relationships between bits
 - C. adding redundancy to data in fixed-size blocks**
 - D. regulating data flow
-

Q.4 Hamming distance is important for error detection in coding as

- A. it measures the signal-to-noise ratio in communication.
 - B. it represents the number of corrupted bits during transmission.**
 - C. it indicates the strength of the error correction code.
 - D. it measures the transmission speed of data.
-

Q.5 If the Hamming distance between sent and received code word is _____, then it shows that received data is corrupted.

- A. =1
 - B. $\neq 1$
 - C. $\neq 0$**
 - D. = 0
-

Q.6 Primary purpose of adding redundant bits to transmitted data in error detection and correction is _____

- A. to reduce network congestion
 - B. to enhance data encryption
 - C. to increase data transmission speed
 - D. to detect or correct errors**
-

Q.7 Redundancy contribute to error detection by _____

- A. reducing the size of data frames
 - B. increasing the speed of data transmission
 - C. adding extra bits to create relationships for error detection**
 - D. encrypting data for secure transmission
-

Q.8 In character oriented protocol, each character of frame is encoded using----- coding system.

- A. NRZ
 - B. Unicode
 - C. Bipolar AMI
 - D. ASCII**
-

Q.9 The term "single-bit error" refer to _____

- A. changing the shape of the signal
 - B. interfering with the flow control process
 - C. changing multiple bits in a data unit
 - D. changing 1 bit of a data unit**
-

Q.10 In digital communication, why do we use a checksum for error detection in a message?

- A. To encrypt the data
 - B. To facilitate faster data transmission
 - C. To correct errors
 - D. To ensure even or odd parity**
-

Q.1 Hamming distance is important for error detection in coding as

- A. it indicates the strength of the error correction code.
 - B. it measures the transmission speed of data.
 - C. it represents the number of corrupted bits during transmission.
 - D. it measures the signal-to-noise ratio in communication.
-

Q.2 The Hamming distance measure _____ in the context of error control coding

- A. the frequency of data errors
 - B. the strength of encryption
 - C. the number of differences between two codewords
 - D. the time taken for data transmission
-

Q.3 _____ is not a function of Data Link Control (DLC).

- A. Error control
 - B. Routing
 - C. Framing
 - D. Flow control
-

Q.4 In _____ type of error, two or more bits in the data unit will be changed.

- A. Mono
 - B. Zero
 - C. Single bit
 - D. Burst
-

Q.5 The term "single-bit error" refer to _____

- A. interfering with the flow control process
 - B. changing multiple bits in a data unit
 - C. changing the shape of the signal
 - D. changing 1 bit of a data unit
-

Q.6 Stop-and-Wait protocol is used for _____.

- A. Neither flow nor error control
 - B. Error control only
 - C. Flow control only
 - D. Both flow and error control
-

Q.7 _____ is the Hamming distance between the binary strings '01001' and '10001.'

- A. 3
 - B. 0
 - C. 1
 - D. 2**
-

Q.8 Checksum is an error-detection technique that can be applied to a message of ____ length.

- A. Exactly 4
 - B. Any**
 - C. Exactly 8
 - D. Exactly 32
-

Q.9 What is the primary function of a checksum in digital communication?

- A. Data compression
 - B. Error correction
 - C. Encryption
 - D. Error detection**
-

Q.10 In block coding, how is the length of codewords (n) related to the length of data words (k)?

- A. $n = k$
 - B. $n > k$**
 - C. $n < k$
 - D. $n = 2k$
-

Q.1 The flag in PPP is a byte that needs to be _____ whenever it appears in the data section of the frame.

- A. Cleaned
- B. Duplicated
- C. Blocked
- D. Escaped**

Q.2 Checksum is an error-detection technique that can be applied to a message of ____ length.

- A. Any**
- B. Exactly 4
- C. Exactly 32
- D. Exactly 8

Q.3 If we need to correct a single error in an 8-bit data unit, we need to consider _____ possible error locations.

- A. 8**
- B. 2
- C. 4
- D. 16

Q.4 In Block coding we divide our message into blocks of k bits called as _____

- A. Passwords
- B. Cross words
- C. Data words**
- D. Code words

Q.5 PAP and CHAP are used by _____ as Authentication Protocols.

- A. SMTP
- B. ICMP
- C. TCP/ IP
- D. PPP**

Q.6 In block coding, how is the length of codewords (n) related to the length of data words (k)?

- A. $n = k$
- B. $n > k$**
- C. $n = 2k$
- D. $n < k$

Q.7 _____ is the process of converting digital data to digital signals.

- A. CSMA/CA
 - B. Line coding**
 - C. Segmentation
 - D. CSMA/CD
-

Q.8 One of the most common protocols for point-to-point access is _____.

- A. TCP
 - B. IP
 - C. UDP
 - D. PPP**
-

Q.9 Which of the following is a subset of Cyclic Codes?

- A. Parity Check
 - B. Hamming Codes**
 - C. Cyclic Redundancy Check (CRC)
 - D. Linear Block Codes
-

Q.10 In order to find the location of two errors in an eight bit stream, we have to see ____ different combinations.

- A. 28**
 - B. 16
 - C. 4
 - D. 8
-